In [138…
```
import os
import sys
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print ("all modules imported successfully")
```

```
all modules imported successfully
```

In [139…
```
df = pd.read_csv('tested.csv')
df.head()
```

Out[139…

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	892	0	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292
1	893	1	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000
2	894	0	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875
3	895	0	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625
4	896	1	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875

In [140…
```
print("First 5 Rows:")
display(df.head())

print("Dataset Shape:")
print(df.shape)

print("Dataset Information:")
df.info()

print("Statistical Summary:")
display(df.describe())
```

```
First 5 Rows:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	892	0	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292
1	893	1	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000
2	894	0	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875
3	895	0	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625
4	896	1	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875

```
Dataset Shape:
(418, 10)
Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 10 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  418 non-null    int64  
 1   Survived     418 non-null    int64  
 2   Pclass       418 non-null    int64  
 3   Name         418 non-null    str    
 4   Sex          418 non-null    str    
 5   Age          332 non-null    float64
 6   SibSp        418 non-null    int64  
 7   Parch        418 non-null    int64  
 8   Ticket       418 non-null    str    
 9   Fare         417 non-null    float64
dtypes: float64(2), int64(5), str(3)
memory usage: 32.8 KB
Statistical Summary:
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	418.000000	418.000000	418.000000	332.000000	418.000000	418.000000	417.000000
mean	1100.500000	0.363636	2.265550	30.272590	0.447368	0.392344	35.627188
std	120.810458	0.481622	0.841838	14.181209	0.896760	0.981429	55.907576
min	892.000000	0.000000	1.000000	0.170000	0.000000	0.000000	0.000000
25%	996.250000	0.000000	1.000000	21.000000	0.000000	0.000000	7.895800
50%	1100.500000	0.000000	3.000000	27.000000	0.000000	0.000000	14.454200
75%	1204.750000	1.000000	3.000000	39.000000	1.000000	0.000000	31.500000
max	1309.000000	1.000000	3.000000	76.000000	8.000000	9.000000	512.329200

In [141…

```
# Check missing values in each column
print("Missing Values:")
print(df.isnull().sum())

# Check total missing values
print("\nTotal Missing Values:")
print(df.isnull().sum().sum())
```

```
Missing Values:
PassengerId     0
Survived        0
Pclass          0
Name            0
Sex             0
Age            86
SibSp           0
Parch           0
Ticket          0
Fare            1
dtype: int64

Total Missing Values:
87
```

In [142…

```
# Calculate missing values percentage
missing = df.isnull().sum()

missing_percentage = (missing / len(df)) * 100

missing_report = pd.DataFrame({
    "Missing Values": missing,
    "Missing Percentage": missing_percentage
})

display(missing_report)
```

	Missing Values	Missing Percentage
PassengerId	0	0.000000
Survived	0	0.000000
Pclass	0	0.000000
Name	0	0.000000
Sex	0	0.000000
Age	86	20.574163
SibSp	0	0.000000
Parch	0	0.000000
Ticket	0	0.000000
Fare	1	0.239234

In [143…
```
# Fill missing values using median
df["Age"] = df["Age"].fillna(df["Age"].median())
df["Fare"] = df["Fare"].fillna(df["Fare"].median())

# Check missing values again
print(df.isnull().sum())

print("\nTotal Missing Values:",
      df.isnull().sum().sum())
```

```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
dtype: int64

Total Missing Values: 0
```

In [144…
```
print("duplicate rows:", df.duplicated().sum())
```

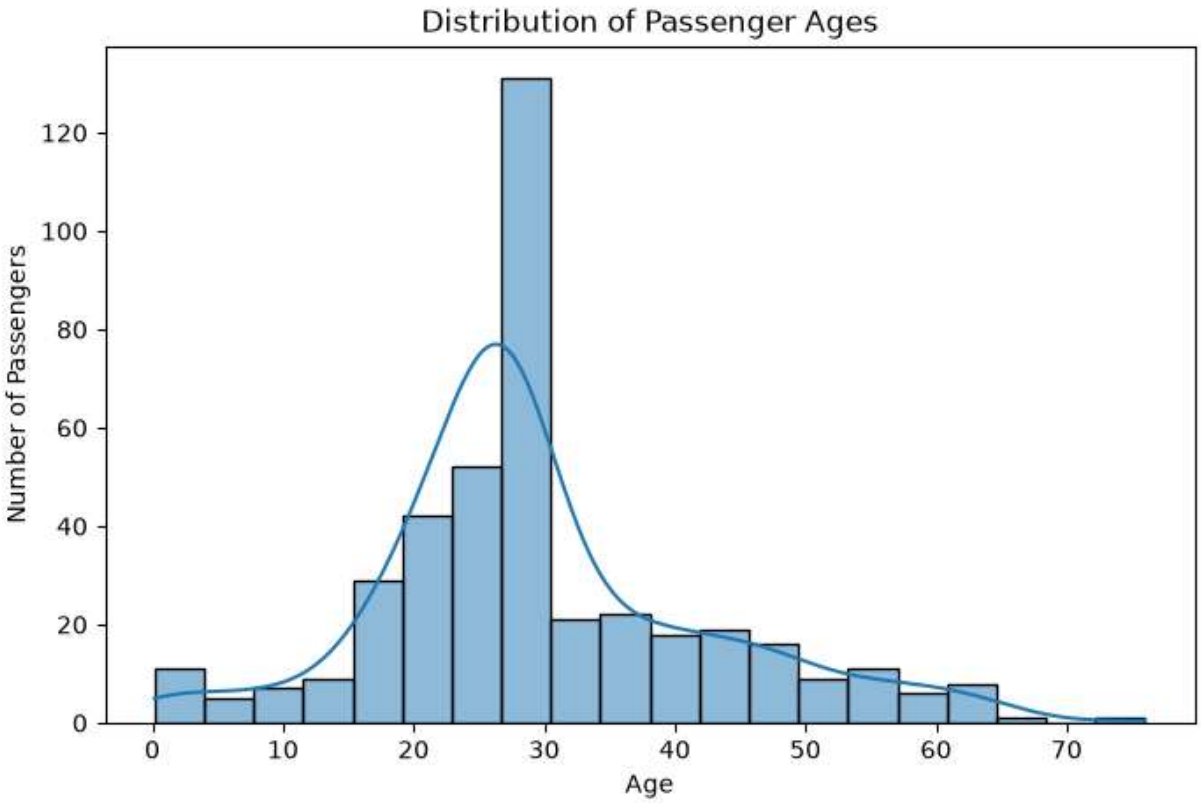
```
duplicate rows: 0
```

In [145…
```
plt.figure(figsize=(8, 5))

sns.histplot(data=df, x="Age", bins=20, kde=True)

plt.title("Distribution of Passenger Ages")
plt.xlabel("Age")
plt.ylabel("Number of Passengers")
```

```
plt.show()
```

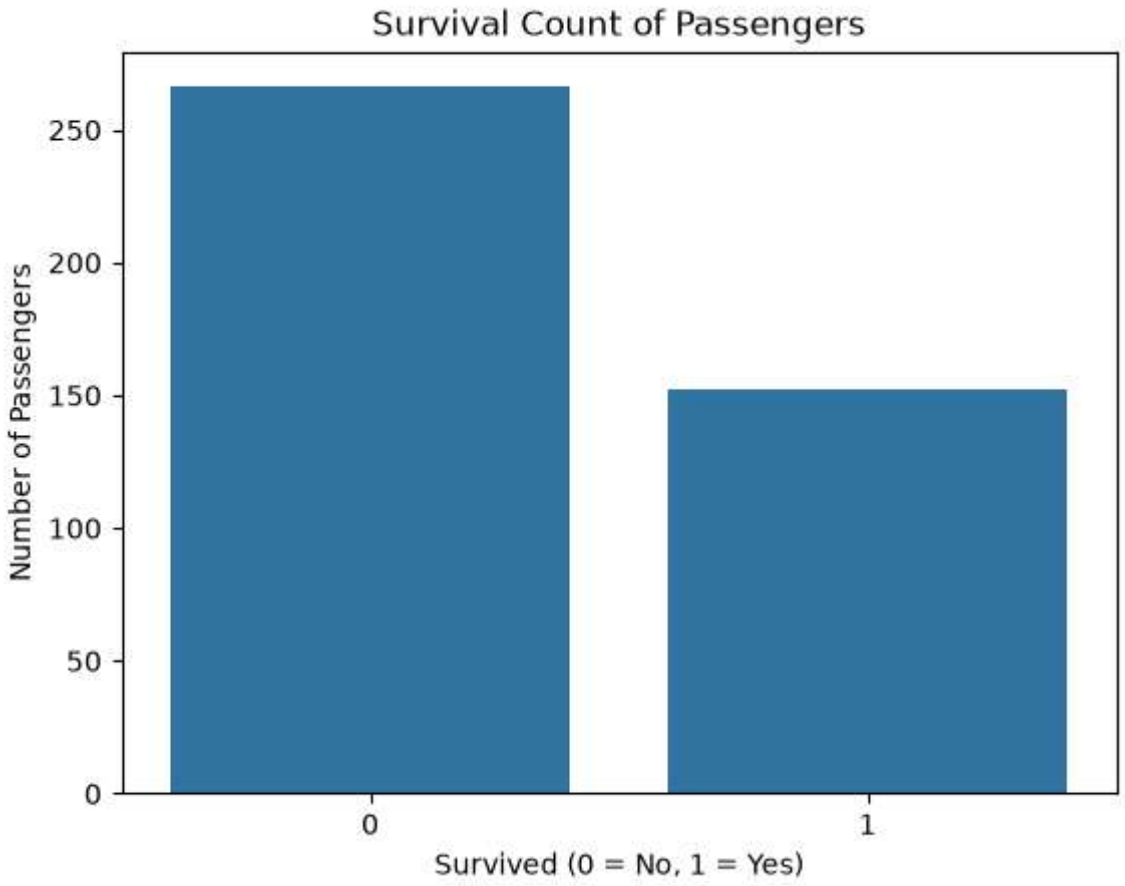


In [146…

```
sns.countplot(data=df, x="Survived")

plt.title("Survival Count of Passengers")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Number of Passengers")

plt.show()
```

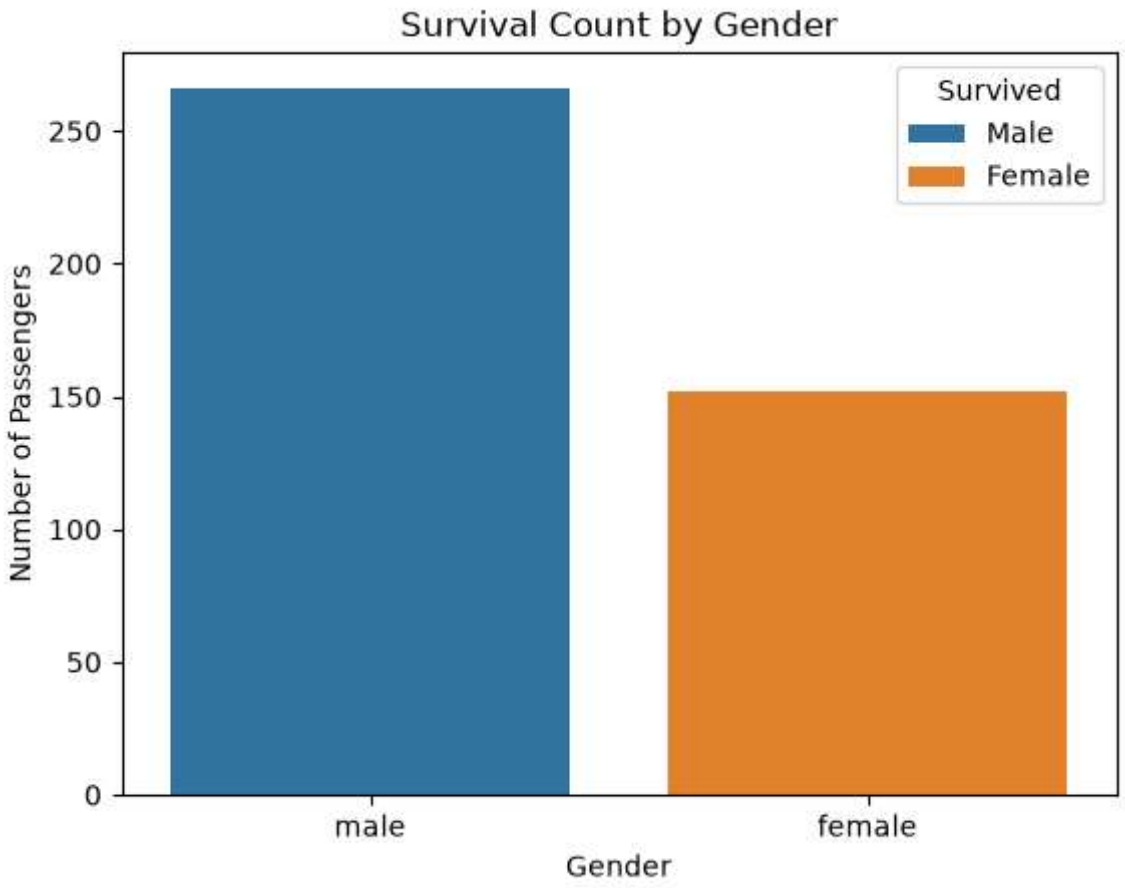


In [147…

```
sns.countplot(data=df, x="Sex", hue="Survived")

plt.title("Survival Count by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")

plt.legend(title="Survived", labels=["Male", "Female"])
plt.show()
```

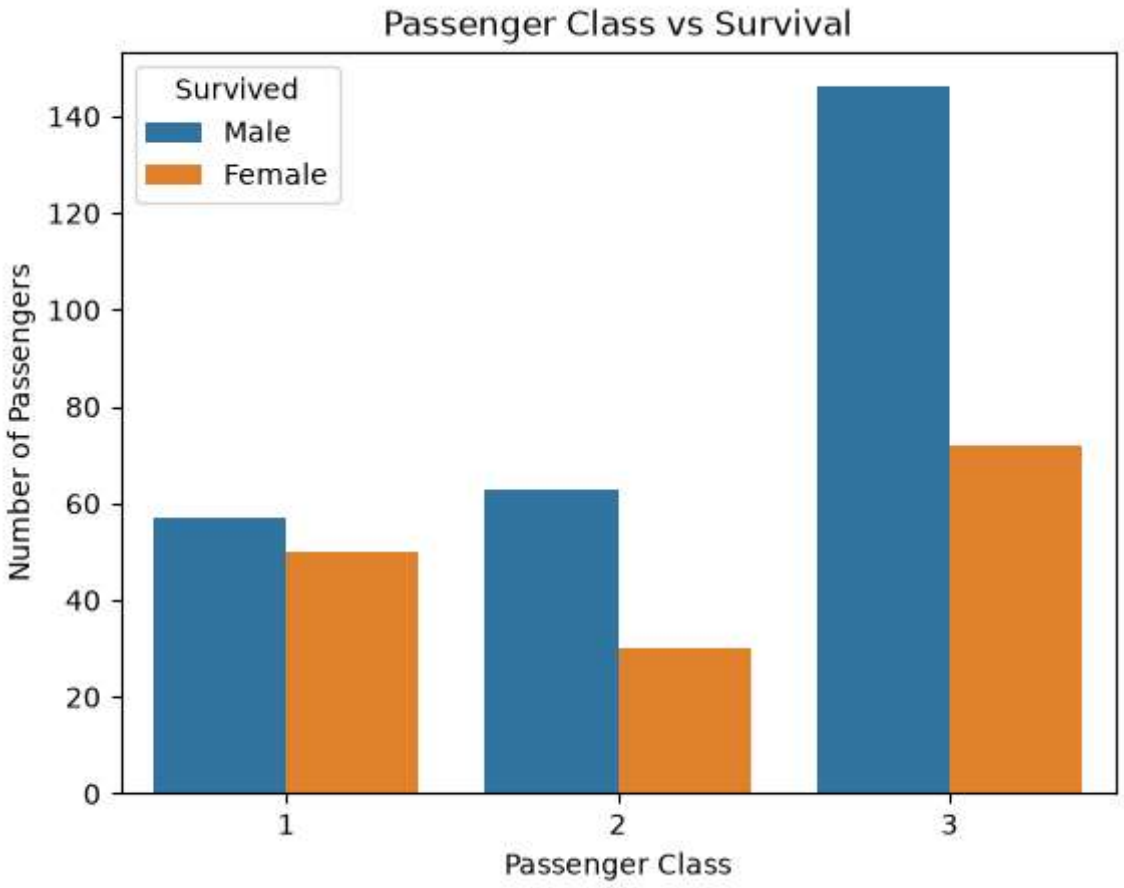


In [148…

```
sns.countplot(data=df, x="Pclass", hue="Survived")

plt.title("Passenger Class vs Survival")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")

plt.legend(title="Survived", labels=["Male", "Female"])
plt.show()
```



In [149…

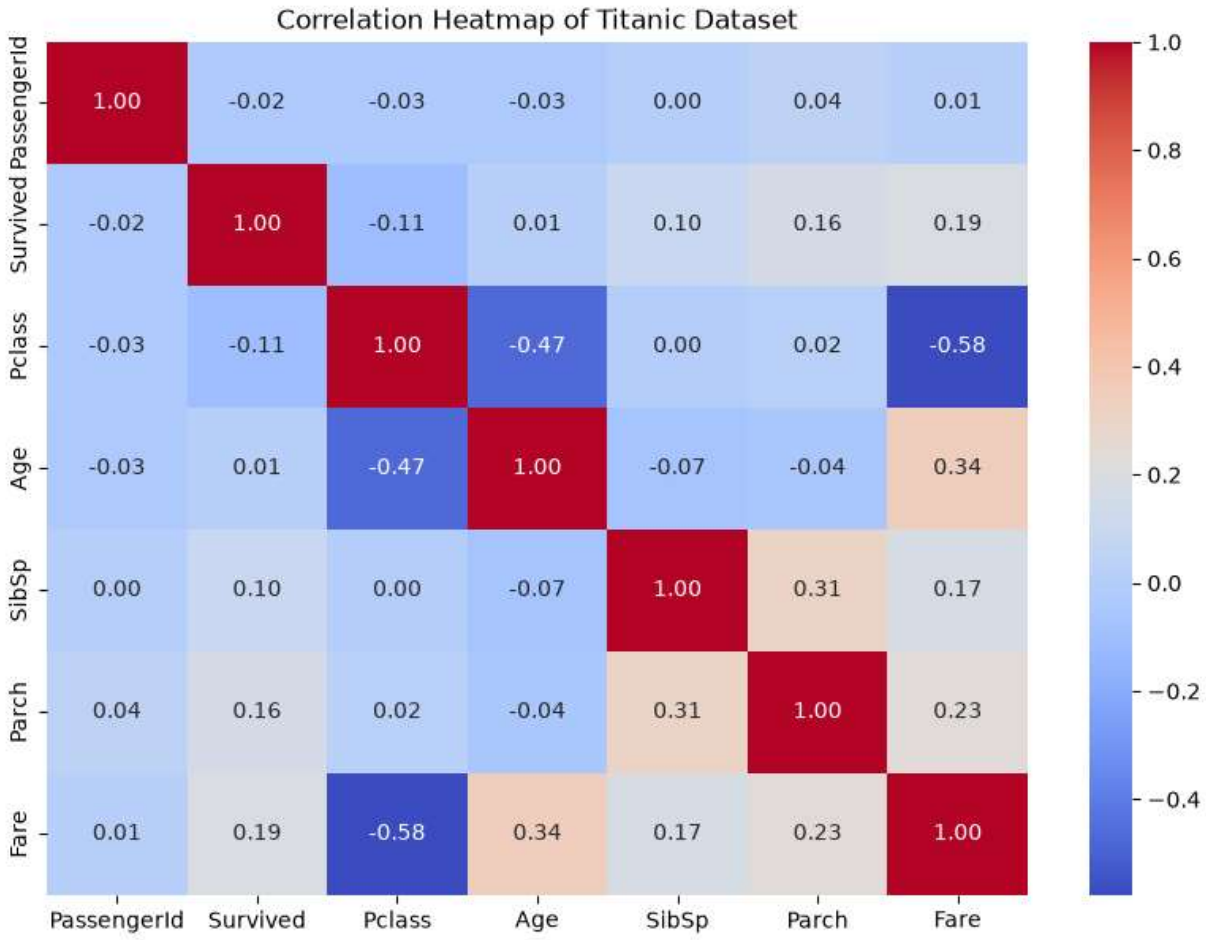
```
# Select numerical columns
numeric_df = df.select_dtypes(include="number")

# Calculate correlation
correlation = numeric_df.corr()

# Create heatmap
plt.figure(figsize=(10, 7))

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Heatmap of Titanic Dataset")
plt.show()
```

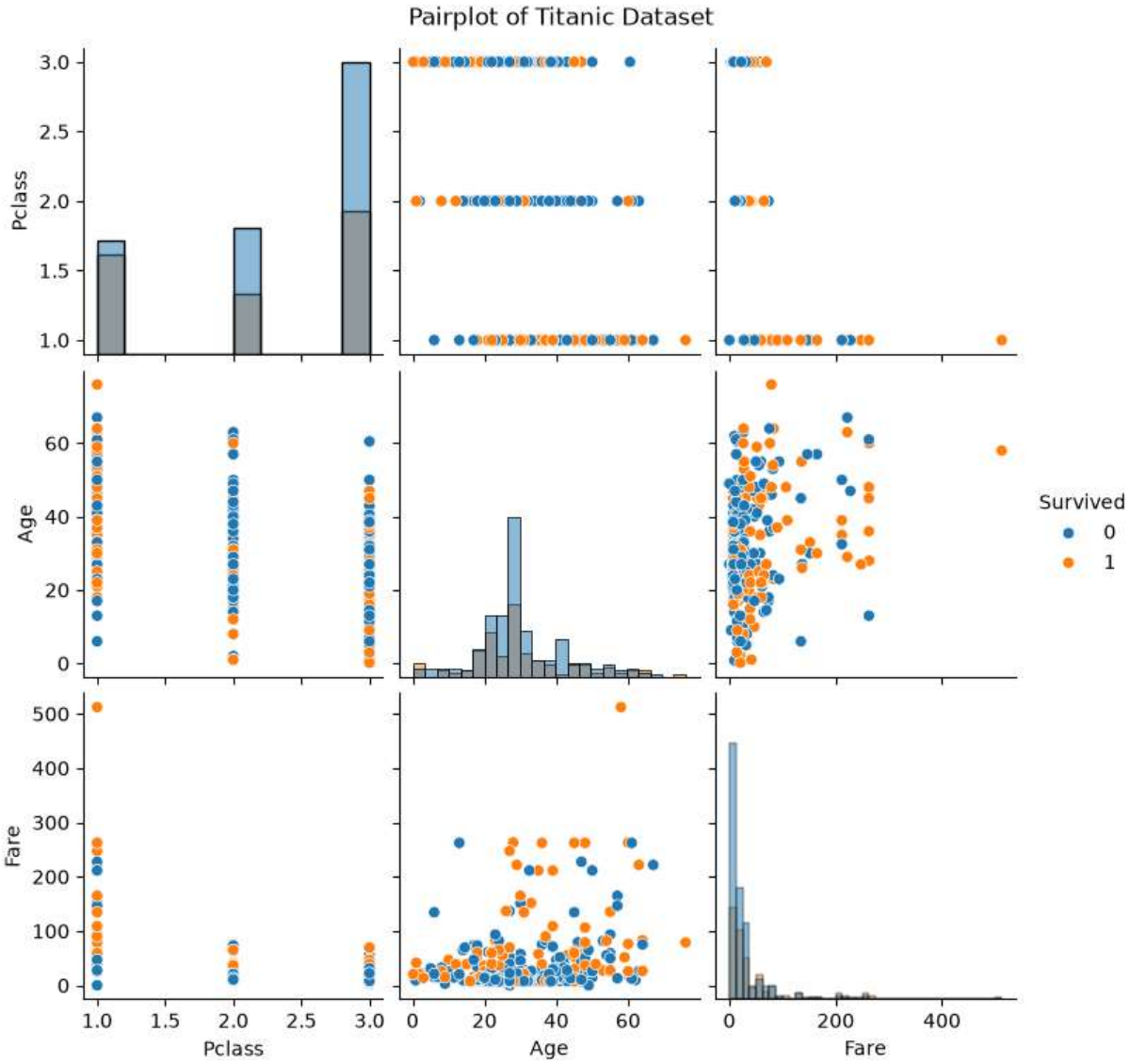


In [150…

```
sns.pairplot(
    df[["Survived", "Pclass", "Age", "Fare"]],
    hue="Survived",
    diag_kind="hist"
)

plt.suptitle(
    "Pairplot of Titanic Dataset",
    y=1.02
)

plt.show()
```



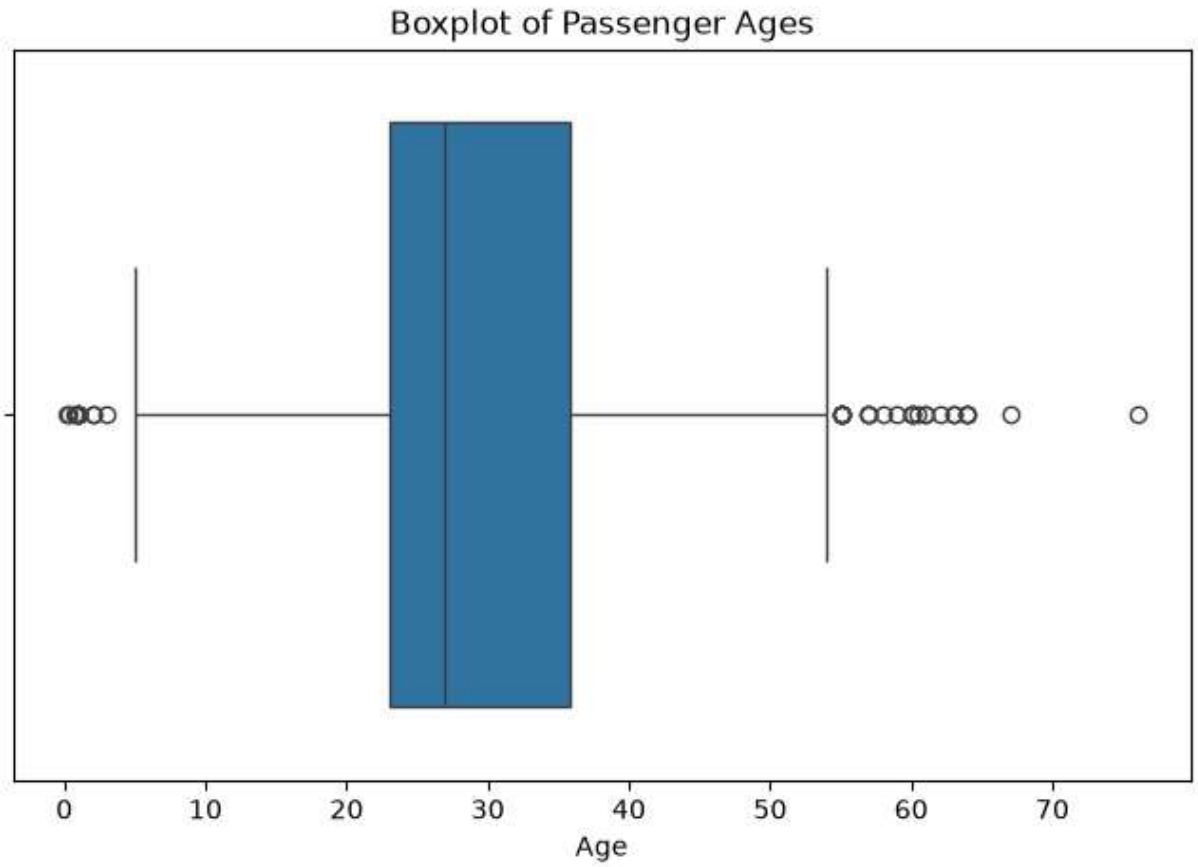
In [151…

```
plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x="Age")

plt.title("Boxplot of Passenger Ages")
plt.xlabel("Age")

plt.show()
```



In [152…

```
print("FINAL EDA SUMMARY")
print("-" * 40)

print("1. Total Passengers:", len(df))

print("\n2. Missing Values:")
print(df.isnull().sum().sum())

print("\n3. Duplicate Rows:")
print(df.duplicated().sum())

print("\n4. Survival Count:")
print(df["Survived"].value_counts())

print("\n5. Survival Percentage:")
print((df["Survived"].value_counts(normalize=True) * 100).round(2))

print("\n6. Average Age:")
print(round(df["Age"].mean(), 2))

print("\n7. Average Fare:")
print(round(df["Fare"].mean(), 2))

print("\n8. Survival by Gender:")
print(df.groupby("Sex")["Survived"].mean().round(2))

print("\n9. Survival by Passenger Class:")
print(df.groupby("Pclass")["Survived"].mean().round(2))
```

```
FINAL EDA SUMMARY
----------------------------------------
1. Total Passengers: 418

2. Missing Values:
0

3. Duplicate Rows:
0

4. Survival Count:
Survived
0    266
1    152
Name: count, dtype: int64

5. Survival Percentage:
Survived
0    63.64
1    36.36
Name: proportion, dtype: float64

6. Average Age:
29.6

7. Average Fare:
35.58

8. Survival by Gender:
Sex
female    1.0
male      0.0
Name: Survived, dtype: float64

9. Survival by Passenger Class:
Pclass
1    0.47
2    0.32
3    0.33
Name: Survived, dtype: float64
```